Surprise and Entropy

Measuring uncertainty in bits

Prof. Nipun Batra

Mathematical Foundations for AI · IIT Gandhinagar

How many yes/no questions?

One bit per character

Model reports and compression benchmarks describe strong language models as reaching roughly one bit per character on English text.

Shannon ran the human version in 1951: readers guessing the next character of English landed between roughly 0.6 and 1.3 bits per character.

Both statements use the **bit** as a unit of uncertainty. Exact figures depend on the text and the tokenizer; the unit is what this lecture defines.

A guessing game makes the unit concrete.

Guess a number from 1 to 8

I choose one number uniformly from $1, \ldots, 8$. You may ask yes/no questions.

A good first question splits the possibilities in half:

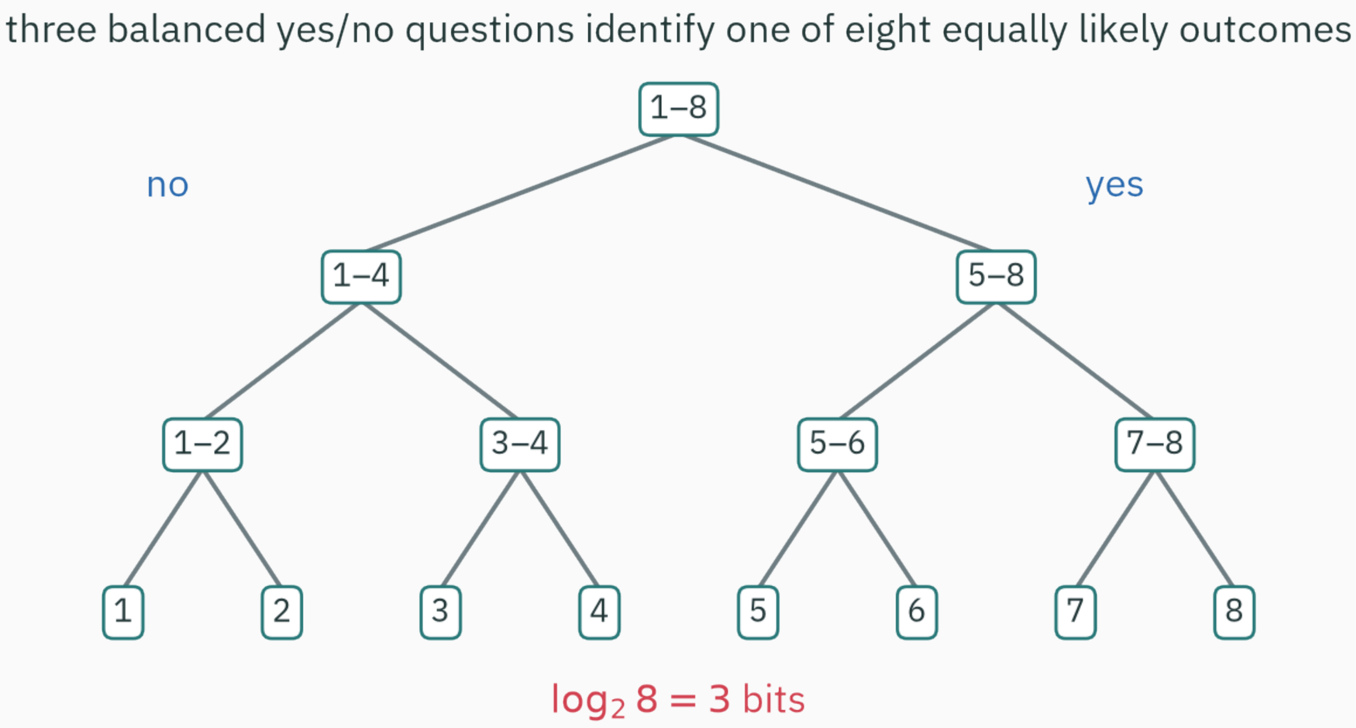
“Is the number greater than 4?”

Then split the remaining four into two, and the remaining two into one.

Three balanced questions always suffice.

A balanced question tree

VISUAL



One yes/no answer carries at most one bit. Distinguishing eight equally likely outcomes takes $\log_2 8 = 3$ bits.

Why balanced questions work

After one balanced answer, 8 possibilities become 4.

After two answers, 4 become 2.

After three answers, 2 become 1.

$8 \to 4 \to 2 \to 1.$

Each answer halves the remaining uncertainty.

Sixteen outcomes

If the outcomes are equally likely, the same reasoning gives

$$16 \to 8 \to 4 \to 2 \to 1.$$

Four binary questions are needed:

$$\log_2 16 = 4.$$

For N equally likely outcomes, the ideal number is $\log_2 N$ bits.

A balance puzzle with twelve coins

Twelve coins look identical; exactly one is counterfeit, either heavier or lighter than the rest. A balance scale compares any two groups of coins.

Each weighing returns one of **three** results: left heavy, right heavy, balanced.

Possible states: $12 \text{ coins} \times 2 \text{ directions} = 24.$

How many weighings identify the coin and its direction?

Count the weighings

w weighings produce at most 3^w distinct result sequences, so $24 \leq 3^w$ is required.

$$3^2 = 9 < 24 \leq 27 = 3^3 \quad \Rightarrow \quad w \geq 3.$$

Three weighings also suffice — MacKay (Ch 4) gives a strategy whose weighings keep the three results near-equally likely.

The same count in bits: the answer holds $\log_2 24 \approx 4.58$ bits, one weighing yields at most $\log_2 3 \approx 1.58$ bits, and $\frac{\log_2 24}{\log_2 3} = \log_3 24 \approx 2.89$.

Outcomes need not be equally likely

Suppose tomorrow's weather probabilities are

$$P(\text{sun}) = 0.75, \quad P(\text{cloud}) = 0.20, \quad P(\text{rain}) = 0.05.$$

"Sun" should be easy to identify because it happens often. "Rain" can receive a longer description because it is rare.

Information theory assigns a numerical surprise to each outcome before averaging over them.

Learning outcomes

By the end of this lecture you will be able to:

1. derive the form $-\log p$ from basic requirements,
2. compute self-information in bits and nats,
3. compute entropy for a finite distribution,
4. interpret entropy as average surprise and ideal average question count,
5. analyze the entropy of a biased coin,
6. derive ⭐ the maximum-entropy distribution on a finite set,
7. and estimate entropy from observed frequencies.

Surprise of one outcome

Three requirements

Let $I(p)$ denote the information gained when an event of probability p occurs.

We want:

1. **Rarer events are more informative:** if $p < q$, then $I(p) > I(q)$.
2. **A certain event gives no information:** $I(1) = 0$.
3. **Independent surprises add:** $I(pq) = I(p) + I(q)$.

Why independent surprises should add

For two independent fair coin flips,

$$P(\mathrm{HH}) = \frac{1}{2} \times \frac{1}{2} = \frac{1}{4}.$$

Learning the first result takes one bit; learning the second takes one more bit.

So learning HH should take two bits:

$$I\left(\frac{1}{4}\right) = I\left(\frac{1}{2}\right) + I\left(\frac{1}{2}\right).$$

The logarithm is forced by addition

The logarithm turns products into sums:

$\log(pq) = \log p + \log q.$

Since $\log p \le 0$ for $0 < p \le 1$, add a minus sign so rarer events receive larger positive values:

$$\boxed{I(p) = -\log p.}$$

Under mild regularity assumptions, the three requirements determine this form up to the choice of log base.

Bits and nats

Base 2 measures information in **bits**:

$$I_2(p) = -\log_2 p.$$

Base e measures information in **nats**:

$$I_e(p) = -\ln p.$$

Conversion:

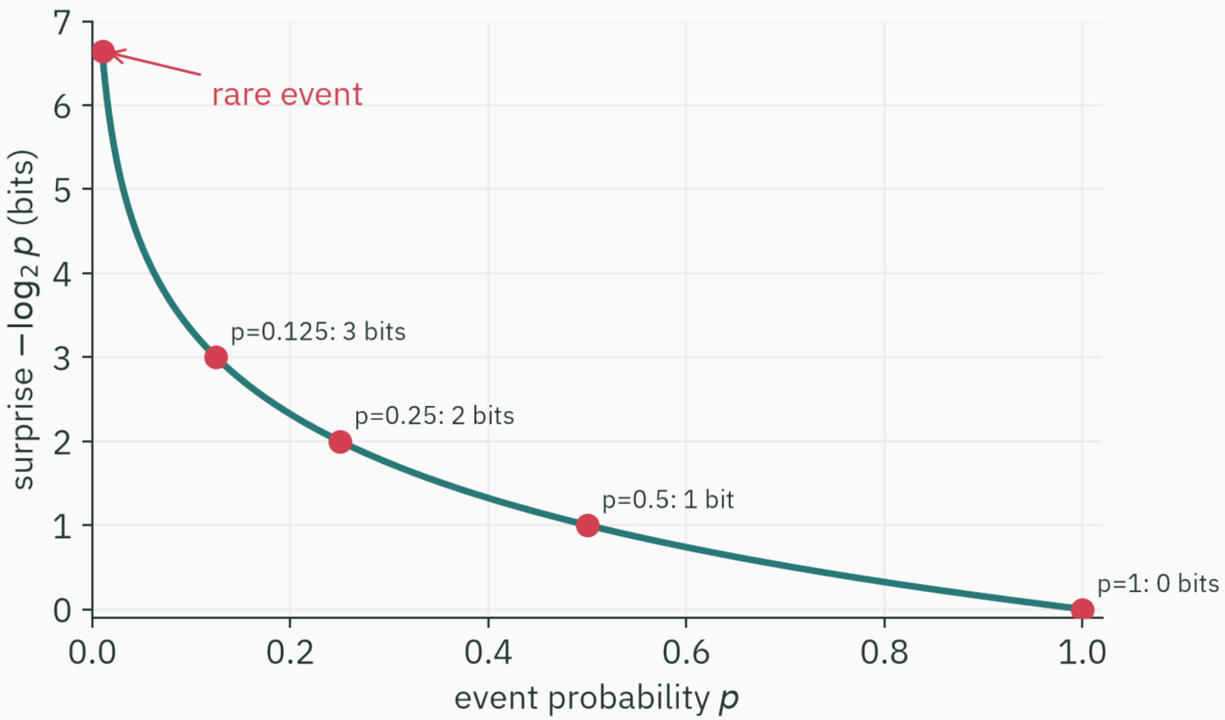
$$1 \quad \text{nat} = \frac{1}{\ln 2} \quad \text{bits} \approx 1.443 \quad \text{bits}.$$

Familiar probabilities

p	$-\log_2 p$	interpretation
1	0 bits	certain
$\frac{1}{2}$	1 bit	one fair binary choice
$\frac{1}{4}$	2 bits	two fair binary choices
$\frac{1}{8}$	3 bits	three fair binary choices
$\frac{1}{1024}$	10 bits	one of 2^{10} outcomes

Surprise as a curve

V VISUAL



Surprise grows without bound as probability approaches zero.

Weather outcomes

For

$$P(\text{sun}) = 0.75, \quad P(\text{cloud}) = 0.20, \quad P(\text{rain}) = 0.05,$$

the surprises are

$$I(\text{sun}) = -\log_2 0.75 \approx 0.415 \quad \text{bits},$$

$$I(\text{cloud}) = -\log_2 0.20 \approx 2.322 \quad \text{bits},$$

$$I(\text{rain}) = -\log_2 0.05 \approx 4.322 \quad \text{bits}.$$

Rain is about ten times more surprising than sun on this scale.

Battleship on an 8-by-8 board

A submarine occupies one of 64 squares, uniformly at random. Each turn you name one square and hear “hit” or “miss”.

On the first guess, $P(\mathrm{hit}) = \frac{1}{64}$ and $P(\mathrm{miss}) = \frac{63}{64}$:

$$I(\mathrm{hit}) = \log_2 64 = 6 \quad \mathrm{bits}, \quad I(\mathrm{miss}) = -\log_2\left(\frac{63}{64}\right) \approx 0.023 \quad \mathrm{bits}.$$

The rare answer is the informative one: a miss barely narrows the board; a hit ends the search.

Every route to the submarine totals six bits

Suppose the first 32 guesses miss and the next one hits.

$$\underbrace{\log_2\left(\frac{64}{63}\right) + \log_2\left(\frac{63}{62}\right) + \ldots + \log_2\left(\frac{33}{32}\right)}_{32 \text{ misses — the product telescopes}} = \log_2\left(\frac{64}{32}\right) = 1 \quad \text{bit.}$$

The hit, with 32 squares left, has probability $\frac{1}{32}$ and adds $\log_2 32 = 5$ bits.

Total $1 + 5 = 6 = \log_2 64$ bits. Multiplying the answer probabilities telescopes the same way, so surprise adds to the same total along any route to the submarine (MacKay, Ch 4).

Log-likelihood already used this quantity

For an observed outcome x under model q,

$$-\log q(x)$$

is both:

1. the surprise assigned by q to x, and
2. the negative log-likelihood contribution of x.

L14 minimized sums of these terms. L24 will interpret that sum as a coding cost.

Question: two independent events

Q QUESTION

Independent events have probabilities $\frac{1}{4}$ and $\frac{1}{8}$. What is the surprise of both occurring?

A. 3 bits

B. 5 bits

C. 12 bits

D. 32 bits

Answer

A ANSWER

Correct: B — 5 bits

Why: $I\left(\left(\frac{1}{4}\right)\left(\frac{1}{8}\right)\right) = I\left(\frac{1}{4}\right) + I\left(\frac{1}{8}\right) = 2 + 3 = 5$ bits.

Average surprise of a source

From one outcome to a distribution

Let a discrete random variable X have outcomes $x_1, \ldots, x_K$ with probabilities $p_1, \ldots, p_K$.

Outcome x_i carries surprise

$$-\log_2 p_i.$$

Before observing X, its expected surprise is

$$\boxed{H(X) = -\sum_{i=1}^{K} p_i \log_2 p_i.}$$

This is Shannon entropy, measured in bits.

Entropy is an expectation

$H(X) = \mathbb{E}[-\log_2 p(X)].$

The random quantity is the surprise of whichever outcome occurs.

High-probability outcomes receive larger weights in this average, even though each carries less surprise individually.

Fair coin

$$P(H) = P(T) = \frac{1}{2}.$$

$$H(X) = -\frac{1}{2}\log_2\left(\frac{1}{2}\right) - \frac{1}{2}\log_2\left(\frac{1}{2}\right).$$

Since $\log_2\left(\frac{1}{2}\right) = -1$,

$$H(X) = \frac{1}{2} + \frac{1}{2} = 1 \quad \text{bit.}$$

One fair flip produces one bit of uncertainty.

A biased coin

Let $P(H) = 0.9$ and $P(T) = 0.1$.

$H(X) = -0.9 \log_2 0.9 - 0.1 \log_2 0.1.$

$H(X) \approx 0.137 + 0.332 = 0.469 \quad \text{bits.}$

The outcome is not known in advance, but it is much more predictable than a fair coin.

A certain coin

If $P(H) = 1$ and $P(T) = 0$,

$$H(X) = -(1 \log_2 1 + 0 \log_2 0) = 0.$$

We use the continuous limit

$$\lim_{p \to 0^+} p \log p = 0.$$

A deterministic source has zero entropy.

Fair die

For six equally likely faces,

$$H(X) = -\sum_{i=1}^{6} \frac{1}{6} \log_2\left(\frac{1}{6}\right).$$

$$H(X) = \log_2 6 \approx 2.585 \quad \text{bits.}$$

The value need not be an integer: entropy is an ideal **average** over many outcomes.

Weather entropy

For probabilities $(0.75, 0.20, 0.05)$,

$$H(X) = 0.75(0.415) + 0.20(2.322) + 0.05(4.322).$$

$$H(X) \approx 0.3113 + 0.4644 + 0.2161 \approx 0.992 \quad \text{bits.}$$

The three-outcome source has less than one bit of average uncertainty because "sun" dominates.

Entropy and average questions

If outcomes are equally likely, a balanced tree uses exactly $\log_2 K$ questions when K is a power of two.

For unequal probabilities, use shorter paths for common outcomes and longer paths for rare ones.

Entropy is the theoretical lower limit on average binary description length, up to coding details developed in L23.

How uncertainty changes with bias

The binary entropy function

For a Bernoulli random variable with $P(X = 1) = p$,

$$H_2(p) = -p\log_2 p - (1-p)\log_2(1-p).$$

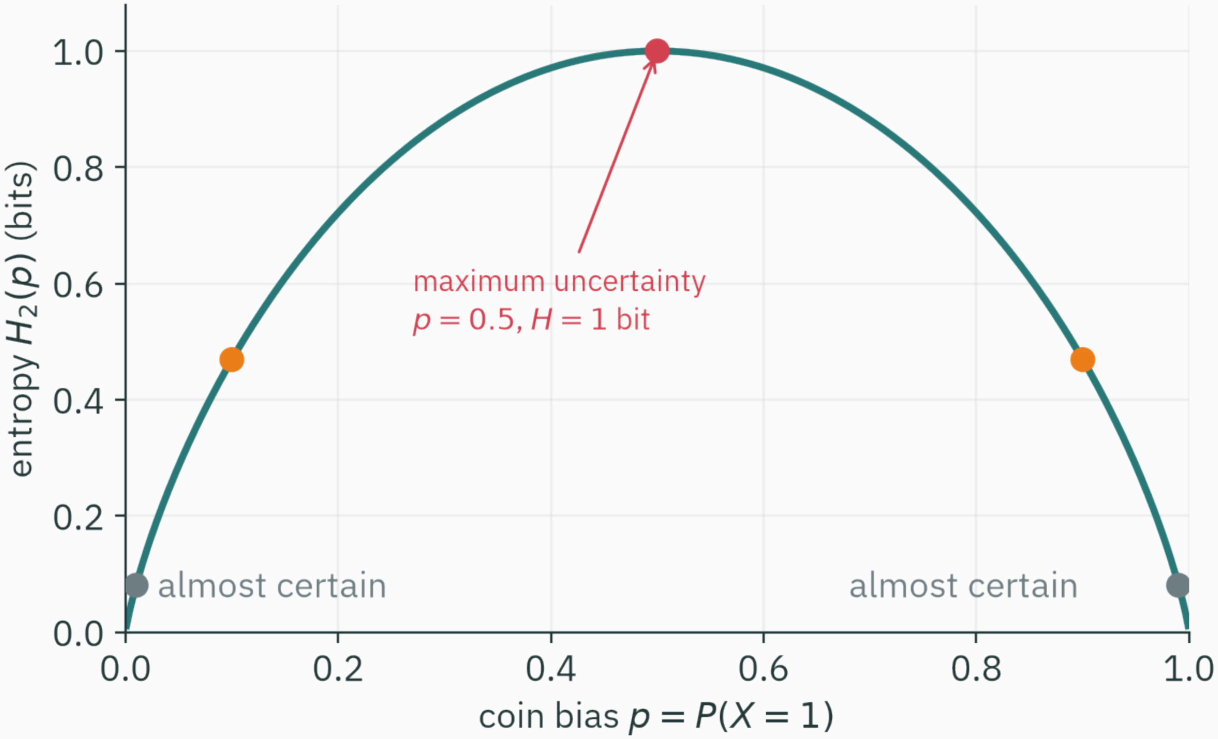
It satisfies

$$H_2(p) = H_2(1-p),$$

because renaming heads and tails does not change uncertainty.

The complete curve

V VISUAL



Entropy is maximal at $p = \frac{1}{2}$ and approaches zero as p approaches 0 or 1.

Interactive: trace the entropy curve

I INTERACTIVE

INTERACTIVE

↗ nipunbatra.github.io/interactive-articles/info-theory

The **info-theory** article recomputes entropy as you drag probability bars. Set two outcomes, drag p from 0.5 toward 0.99, and watch $H_2(p)$ slide down today's curve.

Differentiate to find the maximum

It is cleaner to use natural logs and convert at the end:

$$H_2(p) = -\frac{p \ln p + (1-p)\ln(1-p)}{\ln 2}.$$

$$H_2'(p) = \frac{\ln(1-p) - \ln p}{\ln 2}.$$

Set $H_2'(p) = 0$:

$$\ln(1-p) = \ln p \quad \Rightarrow \quad p = \frac{1}{2}.$$

Check the curvature

$$H_2''(p) = -\frac{1}{\ln 2}\left(\frac{1}{p} + \frac{1}{1-p}\right) < 0$$

for $0 < p < 1$.

So the stationary point $p = \frac{1}{2}$ is the unique maximum.

A fair coin is the most uncertain Bernoulli source.

Ten flips

For independent flips $X_1, ..., X_{10}$,

$$H(X_1, ..., X_{10}) = \sum_{i=1}^{10} H(X_i).$$

1. fair coin: $10 \times 1 = 10$ bits,
2. coin with $p = 0.9$: $10 \times 0.469 = 4.69$ bits.

Independence makes joint entropy additive, matching the additive-surprise requirement.

Question: compare two coins

Q QUESTION

Which Bernoulli source has greater entropy?

A. $p = 0.01$

B. $p = 0.20$

C. $p = 0.50$

D. $p = 0.99$

Answer

ANSWER

Correct: C — $p = 0.50$

Why: The binary entropy curve reaches its unique maximum of one bit at the fair coin.

The least committed distribution

Fixed number of outcomes

Among all distributions over K outcomes, which one has the largest entropy?

$$\max_{\boldsymbol{p}} \quad -\sum_i p_i \ln p_i$$

$$\text{subject to} \quad \sum_i p_i = 1, \quad p_i \geq 0.$$

L19 solved the $K = 3$ version. We now complete the general derivation.

⭐ Form the Lagrangian

D DERIVATION

Assume an interior solution with $p_i > 0$:

$$\mathcal{L}(\boldsymbol{p}, \lambda) = -\sum_i p_i \ln p_i - \lambda\left(\sum_i p_i - 1\right).$$

For each coordinate,

$$\frac{\partial \mathcal{L}}{\partial p_i} = -(\ln p_i + 1) - \lambda = 0.$$

⭐ Every probability is equal

D DERIVATION

Stationarity gives

$$\ln p_i = -1 - \lambda$$

and hence

$$p_i = \exp(-1 - \lambda).$$

The right side does not depend on i, so all p_i have the same value c.

⭐ Normalize and evaluate

D DERIVATION

$$\sum_{i=1}^{K} p_i = Kc = 1 \quad \Rightarrow \quad c = \frac{1}{K}.$$

The entropy is

$$H = -\sum_{i=1}^{K} \frac{1}{K} \log_2\left(\frac{1}{K}\right) = \log_2 K.$$

$H(X) \le \log_2 K$**, with equality for the uniform distribution.**

The Hessian of the objective $-\sum_i p_i \ln p_i$ is diagonal with entries $-\frac{1}{p_i} < 0$ in the simplex interior, so entropy is strictly concave. Therefore this stationary point is the unique global maximum; boundary values follow by continuity.

⭐ The multiplier is a sensitivity

D DERIVATION

At the optimum, $p_i = \exp(-1-\lambda) = \frac{1}{K}$ gives

$\lambda = \ln K - 1.$

L19's reading of λ: relax the constraint to $\sum_i p_i = c$. The optimum becomes $p_i = \frac{c}{K}$, with value

$$H^*(c) = -K \cdot \frac{c}{K} \ln\left(\frac{c}{K}\right) = -c \ln\left(\frac{c}{K}\right).$$

$$\frac{\mathrm{d}H^*}{\mathrm{d}c} = -\ln\left(\frac{c}{K}\right) - 1, \quad \text{at} \quad c = 1: \quad \frac{\mathrm{d}H^*}{\mathrm{d}c} = \ln K - 1 = \lambda.$$

As in L19, the multiplier measures how the optimal value responds when the constraint moves.

Example: four outcomes

distribution	entropy
$(1, 0, 0, 0)$	0 bits
$(0.7, 0.1, 0.1, 0.1)$	1.357 bits
$(0.4, 0.3, 0.2, 0.1)$	1.846 bits
$(0.25, 0.25, 0.25, 0.25)$	**2 bits**

Known constraints change the answer

Uniform is maximum entropy only when normalization is the sole constraint.

If we also know an expected value,

$$\sum_i p_i a_i = \mu,$$

the maximum-entropy solution has exponential form

$$p_i \propto \exp(-\beta a_i).$$

The constraints determine β and the normalizing constant.

Why Gaussians recur

★ OPTIONAL

Among continuous distributions with fixed mean and variance, the Gaussian has maximum differential entropy.

Interpretation: specifying only mean and variance, the Gaussian adds the least extra structure.

This is one reason the Gaussian appeared so often in L12–L15. Differential entropy has subtleties that discrete entropy does not; we use the statement here without proof.

Entropy of an observed source

Empirical probabilities

Given counts $n_1, ..., n_K$ from N observations, estimate

$$\hat{p}_i = \frac{n_i}{N}.$$

The plug-in entropy estimate is

$$\hat{H} = -\sum_i \hat{p}_i \log_2 \hat{p}_i.$$

This is the entropy of the empirical distribution—the MLE from L14 inserted into the entropy formula.

A short symbol sequence

Sequence: `A A A A B B C D`

Counts and probabilities:

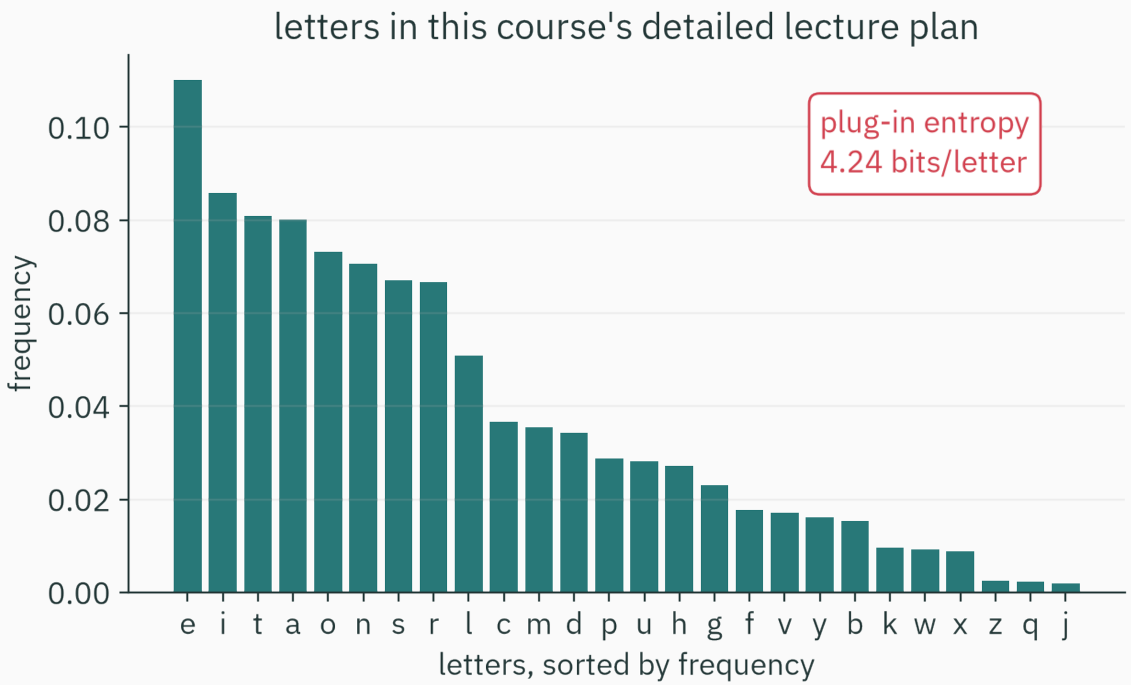
$$\hat{\boldsymbol{p}} = \left(\frac{4}{8}, \frac{2}{8}, \frac{1}{8}, \frac{1}{8}\right).$$

Entropy:

$$\hat{H} = -\left(\frac{1}{2}\right)(-1) - \left(\frac{1}{4}\right)(-2) - 2\left(\frac{1}{8}\right)(-3) = 1.75 \quad \text{bits/symbol.}$$

Course-plan letter frequencies

V VISUAL



The calculation treats letters as independent draws. Natural language has dependencies, so a model that uses context can predict—and compress—better than this unigram estimate.

Compute it directly

```python
from collections import Counter
from math import log2

text = open("lecture-plan-detailed.md").read().lower()
counts = Counter(ch for ch in text if ch.isascii() and ch.isalpha())
N = sum(counts.values())
p = [n/N for n in counts.values()]
H = -sum(pi*log2(pi) for pi in p)
print(H)  # about 4.24 bits/letter
```

Finite-sample caution

The plug-in entropy tends to underestimate the entropy of a large alphabet when data are scarce.

Why?

1. unseen outcomes receive estimated probability zero,
2. rare outcomes have noisy frequency estimates,
3. the logarithm magnifies errors near zero.

For this course, the main lesson is to report the sample size and alphabet along with the estimate.

Independence changes the scale

If letters were independent with entropy 4.24 bits/letter, a 1000-letter text would carry about

$$4240 \quad \text{bits.}$$

But English letters are not independent: after q, the next letter is often u; spaces and words add more structure.

Conditional models reduce uncertainty by using context. Markov chains in L25 make this explicit.

One bit per character, resolved

Treating letters as independent draws gave about 4.24 bits/letter.

Shannon's 1951 prediction experiments: human readers using full context predict English at roughly 0.6 to 1.3 bits per character.

"About one bit per character" for a language model means its average surprise $-\log_2 q(\text{next character})$ on English text is near one bit: context closes most of the gap below the unigram 4.24.

L24 defines this model-based average surprise as cross-entropy.

One scale for uncertainty

Entropy and prediction

Low entropy means a source is predictable:

1. a coin with $p = 0.99$,
2. a sensor stuck at one value,
3. a highly repetitive text.

High entropy means outcomes are spread more evenly:

1. a fair coin,
2. a uniform die,
3. a password drawn uniformly from a large set.

Entropy is not “disorder” by itself

Entropy is defined relative to:

1. a random variable,
2. its possible outcomes,
3. and a probability distribution.

Changing the representation can change the entropy under discussion.

Use the formula and specify the source; do not rely on the word “disorder” alone.

Summary

Information is surprise; entropy is its expected value.

1. Self-information: $I(x) = -\log p(x)$.
2. Independent surprises add because logarithms turn products into sums.
3. Entropy: $H(X) = \mathbb{E}[-\log p(X)]$.
4. A fair Bernoulli source has one bit of entropy.
5. Over K outcomes, the uniform distribution has maximum entropy $\log_2 K$.
6. Empirical entropy is computed from observed frequencies, with finite-sample caution.

Practice

1. Find the surprise of an event with probability $\frac{1}{32}$.
2. Compute the entropy of $(\frac{1}{2}, \frac{1}{4}, \frac{1}{8}, \frac{1}{8})$.
3. Which has higher entropy: $(0.6, 0.4)$ or $(0.9, 0.1)$? Explain without calculation, then check.
4. Find the maximum entropy of a distribution over 100 equally possible labels.

Practice answers

A ANSWER

1. $-\log_2\left(\frac{1}{32}\right) = 5$ bits.
2. $\frac{1}{2}(1) + \frac{1}{4}(2) + 2\left(\frac{1}{8}\right)(3) = 1.75$ bits.
3. $(0.6, 0.4)$ is closer to uniform and has larger entropy: about 0.971 versus 0.469 bits.
4. $\log_2 100 \approx 6.644$ bits, attained by the uniform distribution.

Next: turn probabilities into descriptions

Entropy states an ideal average number of bits. We still need a constructive code.

Next lecture:

1. prefix-free binary codes,
2. Huffman's greedy algorithm,
3. expected code length versus entropy.

Reading: MacKay, *Information Theory, Inference, and Learning Algorithms* — Ch 2 and Ch 4 (guided) · MacKay's video lecture 1 (assigned).

Information is surprise; entropy is its average.